

Calculus 2

3 Lecture 3

Operations with series

Claim 3.1 (The Abel transformation). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences. Then, for every $n, k \in \mathbb{N}$, we have

$$\sum_{i=n+1}^{n+k} a_i b_i = A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i)$$

where

$$A_m = \sum_{j=1}^m a_j.$$

Proof. Note that $a_i = A_i - A_{i-1}$. Hence

$$\begin{aligned} \sum_{i=n+1}^{n+k} a_i b_i &= \sum_{i=n+1}^{n+k} (A_i - A_{i-1}) b_i \\ &= \sum_{i=n+1}^{n+k} A_i b_i - \sum_{i=n+1}^{n+k} A_{i-1} b_i \\ &= \sum_{i=n+1}^{n+k} A_i b_i - \sum_{i=n}^{n+k-1} A_i b_{i+1} \\ &= A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i). \end{aligned}$$

□

Claim 3.2 (The Dirichlet Test). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences of real numbers satisfying the following conditions:

a) there exists $M > 0$ such that

$$\left| \sum_{i=1}^n a_i \right| \leq M \quad \text{for every } n \in \mathbb{N};$$

b) $\{b_i\}_{i=1}^{\infty}$ is monotonic; that is, either

$$b_i \geq b_{i+1} \quad (\forall i \in \mathbb{N})$$

or

$$b_i \leq b_{i+1} \quad (\forall i \in \mathbb{N});$$

c)

$$\lim_{i \rightarrow +\infty} b_i = 0.$$

Then the series

$$\sum_{i=1}^{\infty} a_i b_i$$

converges.

Proof. Due to [Claim 1.2](#) (Cauchy's criterion),

$$\left(\sum_{i=1}^{\infty} a_i b_i \text{ converges} \right) \iff \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall n, k \in \mathbb{N}, n > N(\varepsilon) \implies \left| \sum_{i=n+1}^{n+k} a_i b_i \right| < \varepsilon \right).$$

Thus, it suffices to prove the condition on the right.

Let $\varepsilon > 0$ be arbitrary. Since $\lim_{i \rightarrow +\infty} b_i = 0$, there exists $N_1(\varepsilon) \in \mathbb{N}$ such that

$$\forall n > N_1(\varepsilon), \quad |b_n| < \frac{\varepsilon}{2M}.$$

Set $N(\varepsilon) = N_1(\varepsilon)$. For every $n, k \in \mathbb{N}$ with $n > N(\varepsilon)$, the Abel transformation gives

$$\begin{aligned} \left| \sum_{i=n+1}^{n+k} a_i b_i \right| &= \left| A_{n+k} b_{n+k} - A_n b_{n+1} - \sum_{i=n+1}^{n+k-1} A_i (b_{i+1} - b_i) \right| \\ &\leq |A_{n+k}| |b_{n+k}| + |A_n| |b_{n+1}| + \sum_{i=n+1}^{n+k-1} |A_i| |b_{i+1} - b_i| \\ &\leq M |b_{n+k}| + M |b_{n+1}| + M \sum_{i=n+1}^{n+k-1} |b_{i+1} - b_i|. \end{aligned} \quad (*)$$

Suppose first that $b_i \geq b_{i+1}$ for every $i \in \mathbb{N}$. Together with $\lim b_i = 0$, this implies $b_i \geq 0$ for every $i \in \mathbb{N}$. Therefore, continuing from (*),

$$\begin{aligned} \left| \sum_{i=n+1}^{n+k} a_i b_i \right| &\leq M b_{n+k} + M b_{n+1} + M \sum_{i=n+1}^{n+k-1} (b_i - b_{i+1}) \\ &= M (b_{n+k} + b_{n+1} + b_{n+1} - b_{n+k}) \\ &= 2M b_{n+1} < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon. \end{aligned}$$

Hence, by Cauchy's criterion, $\sum_{i=1}^{\infty} a_i b_i$ converges.

If instead $b_i \leq b_{i+1}$ for every $i \in \mathbb{N}$, then together with $\lim b_i = 0$ this implies $b_i \leq 0$ for every $i \in \mathbb{N}$. The sequence $\{-b_i\}$ is nonincreasing and tends to 0, so by the case already proved, the series $\sum a_i (-b_i)$ converges. Therefore $\sum a_i b_i$ converges as well. $\square$

Claim 3.3 (The Abel Test). Let $\{a_i\}_{i=1}^{\infty}$ and $\{b_i\}_{i=1}^{\infty}$ be two sequences of real numbers satisfying the following conditions:

- a) $\sum_{i=1}^{\infty} a_i$ converges;
- b) $\{b_i\}_{i=1}^{\infty}$ is monotonic;
- c) $\{b_i\}_{i=1}^{\infty}$ is bounded; that is, there exists $L > 0$ such that

$$|b_i| \leq L \quad (\forall i \in \mathbb{N}).$$

Then the series

$$\sum_{i=1}^{\infty} a_i b_i$$

converges.

Proof. Since $\{b_i\}_{i=1}^{\infty}$ is monotonic and bounded, we have (see Calculus 1)

$$\exists \lim_{i \rightarrow +\infty} b_i = b.$$

Note that:

- 1) the sequence $\{b_i - b\}_{i=1}^{\infty}$ satisfies conditions b) and c) of [Claim 3.2](#);
 2) since $\sum_{i=1}^{\infty} a_i$ converges, the sequence of partial sums

$$\left\{ \sum_{k=1}^i a_k \right\}_{i=1}^{\infty}$$

is bounded, so $\{a_i\}$ satisfies condition a) of [Claim 3.2](#).

Therefore, by [Claim 3.2](#), the series

$$\sum_{i=1}^{\infty} a_i(b_i - b)$$

converges. Since

$$\sum_{i=1}^{\infty} a_i b_i = b \sum_{i=1}^{\infty} a_i + \sum_{i=1}^{\infty} a_i(b_i - b),$$

the series $\sum_{i=1}^{\infty} a_i b_i$ converges. □

Claim 3.4 (The Leibniz Test). Let $\{b_i\}_{i=1}^{\infty}$ be a sequence of real numbers satisfying

- a) $\{b_i\}_{i=1}^{\infty}$ is nonincreasing, that is,

$$b_i \geq b_{i+1} \quad (\forall i \in \mathbb{N});$$

- b)

$$\lim_{i \rightarrow +\infty} b_i = 0.$$

Then:

- 1) the series

$$\sum_{i=1}^{\infty} (-1)^{i+1} b_i$$

converges;

- 2) for every $n \in \mathbb{N}$,

$$\left| \sum_{i=n+1}^{\infty} (-1)^{i+1} b_i \right| \leq b_{n+1}.$$

Proof. 1) This follows directly from [Claim 3.2](#) (Dirichlet's test) if we set

$$a_i = (-1)^{i+1},$$

because

$$\left| \sum_{i=1}^n a_i \right| = \left| \sum_{i=1}^n (-1)^{i+1} \right| \leq 1.$$

- 2) Let

$$S_n = \sum_{i=1}^n (-1)^{i+1} b_i \quad (n \in \mathbb{N}).$$

For every $n \in \mathbb{N}$,

$$S_{2n+2} = S_{2n} + b_{2n+1} - b_{2n+2} \geq S_{2n},$$

and

$$S_{2n+1} = S_{2n-1} - b_{2n} + b_{2n+1} \leq S_{2n-1}.$$

Therefore $\{S_{2n}\}_{n=1}^{\infty}$ is nondecreasing and $\{S_{2n-1}\}_{n=1}^{\infty}$ is nonincreasing. Due to [item 1](#)), the sequence $\{S_n\}_{n=1}^{\infty}$ is convergent, so both subsequences converge and have the same limit. Let

$$\lim_{n \rightarrow +\infty} S_n = S.$$

Hence

$$S_{2n} \leq S \leq S_{2n-1} \quad (\forall n \in \mathbb{N}).$$

Consequently,

$$0 \leq S - S_{2n} \leq S_{2n+1} - S_{2n} = b_{2n+1},$$

and

$$0 \leq S_{2n-1} - S \leq S_{2n-1} - S_{2n} = b_{2n}.$$

Thus, for every $n \in \mathbb{N}$,

$$|S - S_n| \leq b_{n+1}.$$

Since

$$S - S_n = \sum_{i=n+1}^{\infty} (-1)^{i+1} b_i,$$

the statement follows. □

Claim 3.5 (Sine and cosine sums). For every $n \in \mathbb{N}$ and $\alpha, b \in \mathbb{R}$, we have

$$\sum_{k=1}^n \sin(\alpha k + b) = \begin{cases} \frac{\sin\left(\frac{n+1}{2}\alpha + b\right) \sin\left(\frac{n}{2}\alpha\right)}{\sin\left(\frac{\alpha}{2}\right)}, & \alpha \notin 2\pi\mathbb{Z}, \\ n \sin b, & \alpha \in 2\pi\mathbb{Z}. \end{cases}$$

Proof. If $\alpha \in 2\pi\mathbb{Z}$, then $\sin(\alpha k + b) = \sin b$, and the formula is immediate.

Assume that $\alpha \notin 2\pi\mathbb{Z}$, and let

$$S_n = \sum_{k=1}^n \sin(\alpha k + b).$$

Using

$$2 \sin x \sin y = \cos(x - y) - \cos(x + y) \quad (\forall x, y \in \mathbb{R}),$$

we obtain

$$\begin{aligned} 2 \sin\left(\frac{\alpha}{2}\right) S_n &= \sum_{k=1}^n 2 \sin(\alpha k + b) \sin\left(\frac{\alpha}{2}\right) \\ &= \sum_{k=1}^n \left(\cos\left(\alpha k + b - \frac{\alpha}{2}\right) - \cos\left(\alpha k + b + \frac{\alpha}{2}\right) \right) \\ &= \cos\left(\alpha + b - \frac{\alpha}{2}\right) - \cos\left(\alpha n + b + \frac{\alpha}{2}\right). \end{aligned}$$

This is a telescoping sum. Now use

$$\cos x - \cos y = 2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{y-x}{2}\right) \quad (\forall x, y \in \mathbb{R})$$

to get

$$2 \sin\left(\frac{\alpha}{2}\right) S_n = 2 \sin\left(\frac{n+1}{2}\alpha + b\right) \sin\left(\frac{n}{2}\alpha\right).$$

It remains to divide both sides by $2 \sin(\alpha/2) \neq 0$. □

Claim 3.6. For every $n \in \mathbb{N}$, we have

1)

$$S_n = \sin 1 + \sin 2 + \cdots + \sin n = \frac{\sin\left(\frac{n+1}{2}\right) \sin\left(\frac{n}{2}\right)}{\sin\left(\frac{1}{2}\right)};$$

2)

$$C_n = \cos 1 + \cos 2 + \cdots + \cos n = \frac{\cos\left(\frac{n+1}{2}\right) \sin\left(\frac{n}{2}\right)}{\sin\left(\frac{1}{2}\right)}.$$

Proof. 1) Follows directly from [Claim 3.5](#) by setting $\alpha = 1$ and $b = 0$.

2) Follows directly from [Claim 3.5](#) by setting $\alpha = 1$ and $b = \pi/2$ and using

$$\sin\left(x + \frac{\pi}{2}\right) = \cos x \quad (\forall x \in \mathbb{R}).$$

□

Definition 3.1. A series $\sum_{k=1}^{\infty} a_k$ is called

- *absolutely convergent* if the series $\sum_{k=1}^{\infty} |a_k|$ converges;
- *conditionally convergent* if it converges but the series $\sum_{k=1}^{\infty} |a_k|$ diverges.

Thus, there are three alternatives for a series $\sum_{k=1}^{\infty} a_k$:

- the series is absolutely convergent;
- the series is conditionally convergent;
- the series is divergent.

Claim 3.7. An absolutely convergent series is convergent. That is,

$$\left(\sum_{k=1}^{\infty} |a_k| \text{ converges}\right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges}\right).$$

Proof. Since

$$0 \leq a_k + |a_k| \leq 2|a_k| \quad (\forall k \in \mathbb{N}),$$

and the series $\sum_{k=1}^{\infty} |a_k|$ converges, due to [Claim 2.4](#) (the First Comparison Test), the series

$$\sum_{k=1}^{\infty} (a_k + |a_k|)$$

converges. Since

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} (a_k + |a_k|) - \sum_{k=1}^{\infty} |a_k|,$$

due to [Claim 1.8](#), the series $\sum_{k=1}^{\infty} a_k$ converges. □

Claim 3.8 (Alternating p -series). The series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$$

1) converges absolutely if $p > 1$;

2) converges conditionally if $0 < p \leq 1$;

3) diverges if $p \leq 0$.

Proof. 1) Since

$$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k^p} \right| = \sum_{k=1}^{\infty} \frac{1}{k^p},$$

item 1) follows from Claim 2.7.

3) If $p \leq 0$, then

$$\lim_{k \rightarrow +\infty} \frac{(-1)^{k+1}}{k^p} \neq 0.$$

Hence item 3) follows from Claim 1.5 (the necessary condition for convergence).

2) If $p > 0$, the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$$

converges due to Claim 3.4 (the Leibniz Test). For $0 < p \leq 1$, the series of absolute values $\sum_{k=1}^{\infty} 1/k^p$ diverges by Claim 2.7. Therefore the convergence is conditional. $\square$

Definition 3.2. Let $\tau : \mathbb{N} \rightarrow \mathbb{N}$ be a *permutation* of the set of natural numbers; that is, τ is a bijection from $\mathbb{N}$ to $\mathbb{N}$. Then the series

$$\sum_{k=1}^{\infty} a_{\tau(k)}$$

is called a *rearrangement* of the series

$$\sum_{k=1}^{\infty} a_k.$$

Roughly speaking, a rearrangement changes the order of the terms of the series. For example,

$$a_2 + a_1 + a_4 + a_3 + a_6 + a_5 + \cdots + a_{2n} + a_{2n-1} + \cdots$$

is a rearrangement of $\sum_{k=1}^{\infty} a_k$.

Claim 3.9 (Cauchy's Rearrangement Theorem). Let

$$\sum_{k=1}^{\infty} a_k$$

be an absolutely convergent series. Then, for any permutation $\tau : \mathbb{N} \rightarrow \mathbb{N}$, we have

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \sum_{k=1}^{\infty} a_k.$$

Proof. Let

$$\sum_{k=1}^{\infty} a_k = \alpha.$$

We need to prove that

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \alpha.$$

By the definition of convergence, it suffices to prove that

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n > N, \quad \left| \alpha - \sum_{k=1}^n a_{\tau(k)} \right| < \varepsilon. \quad (\diamond)$$

Since the series is absolutely convergent, $\sum_{k=1}^{\infty} |a_k|$ converges. Due to [Claim 1.7](#),

$$\forall \varepsilon > 0 \exists M \in \mathbb{N} : \forall n \geq M, \quad \sum_{k=n+1}^{\infty} |a_k| < \varepsilon. \quad (\nabla)$$

Choose $N \in \mathbb{N}$ such that

$$\{1, 2, \dots, M\} \subseteq \{\tau(1), \tau(2), \dots, \tau(N)\}.$$

This can be done since $\tau : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection. Then for every $n > N$,

$$\begin{aligned} \left| \alpha - \sum_{k=1}^n a_{\tau(k)} \right| &= \left| \sum_{k=1}^{\infty} a_k - \sum_{k=1}^n a_{\tau(k)} \right| \\ &\leq \sum_{k=M+1}^{\infty} |a_k| \\ &< \varepsilon, \end{aligned}$$

where the terms $a_1, a_2, \dots, a_M$ cancel due to the choice of N . This proves [\(\diamond\)](#). $\square$

Claim 3.10. Let $\{a_k\}_{k=1}^{\infty}$ be a sequence of real numbers. Let λ_k be the k -th nonnegative term of $\{a_k\}_{k=1}^{\infty}$, and let β_k be the k -th negative term of $\{a_k\}_{k=1}^{\infty}$.

For example, if

$$\{a_k\}_{k=1}^{\infty} = (1, 0, -7, 2, -3, 1, -1, \dots),$$

then

$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = 2, \quad \lambda_4 = 1,$$

and

$$\beta_1 = -7, \quad \beta_2 = -3, \quad \beta_3 = -1.$$

If the series $\sum_{k=1}^{\infty} a_k$ converges conditionally, then:

- 1) there are infinitely many nonnegative and infinitely many negative terms in the sequence $\{a_k\}_{k=1}^{\infty}$;
- 2) both series

$$\sum_{k=1}^{\infty} \lambda_k \quad \text{and} \quad \sum_{k=1}^{\infty} \beta_k$$

are divergent.

Proof. 1) Suppose that the set of nonnegative terms of the sequence $\{a_k\}_{k=1}^{\infty}$ is finite. Then there exists $N \in \mathbb{N}$ such that

$$\forall k > N, \quad a_k < 0.$$

Hence, since $|a_k| = -a_k$ for $k > N$,

$$\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^N |a_k| + \sum_{k=N+1}^{\infty} |a_k| = M - \sum_{k=N+1}^{\infty} a_k,$$

where $M = \sum_{k=1}^N |a_k|$ is a constant. The series on the right converges because $\sum a_k$ converges. Therefore $\sum a_k$ converges absolutely, which is a contradiction. Thus, the set of nonnegative terms is infinite. In a similar way, one proves that the set of negative terms is also infinite.

- 2) Due to [item 1\)](#), we can define the series $\sum_{k=1}^{\infty} \lambda_k$ and $\sum_{k=1}^{\infty} \beta_k$.

Suppose first that both series are convergent:

$$\sum_{k=1}^{\infty} \lambda_k = \lambda \in \mathbb{R}, \quad \sum_{k=1}^{\infty} \beta_k = \beta \in \mathbb{R}.$$

Then, for every $N \in \mathbb{N}$,

$$\sum_{k=1}^N |a_k| \leq \sum_{k=1}^N \lambda_k + \sum_{k=1}^N |\beta_k| \leq \lambda + |\beta|.$$

Therefore, due to [Claim 2.2](#), the series $\sum_{k=1}^{\infty} |a_k|$ converges. This contradicts conditional convergence. Hence at least one of the series $\sum \lambda_k$ or $\sum \beta_k$ diverges.

Suppose that one of these series converges and the other diverges. Without loss of generality, assume that

$$\sum_{k=1}^{\infty} \lambda_k \text{ diverges,} \quad \sum_{k=1}^{\infty} \beta_k = \beta \in \mathbb{R}.$$

Then

$$\forall E > 0 \exists N \in \mathbb{N} : \forall n \geq N, \quad \sum_{k=1}^n \lambda_k > E - \beta.$$

Choose $M \in \mathbb{N}$ such that

$$\{\lambda_1, \dots, \lambda_N\} \subseteq \{a_1, \dots, a_M\}.$$

Then

$$\sum_{k=1}^M a_k \geq \sum_{k=1}^N \lambda_k + \beta > E - \beta + \beta = E.$$

Since $E > 0$ is arbitrary, the partial sums of $\sum a_k$ are unbounded above, so $\sum a_k$ diverges. This is a contradiction. Therefore both $\sum \lambda_k$ and $\sum \beta_k$ are divergent. $\square$

Claim 3.11 (The Riemann Rearrangement Theorem). Let

$$\sum_{k=1}^{\infty} a_k$$

be a conditionally convergent series. Then, for any $\alpha \in \mathbb{R}$, there exists a permutation $\tau : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\sum_{k=1}^{\infty} a_{\tau(k)} = \alpha.$$

Proof. Let λ_k be the k -th nonnegative term and β_k the k -th negative term of the sequence $\{a_k\}_{k=1}^{\infty}$. We construct the rearranged series $\sum_{k=1}^{\infty} a_{\tau(k)}$ as follows.

Let $\alpha \in \mathbb{R}$ be arbitrary.

- Let p_1 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k > \alpha.$$

Such p_1 exists because $\sum_{k=1}^{\infty} \lambda_k = +\infty$ by [item 2\)](#) of [Claim 3.10](#).

- Let q_1 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k < \alpha.$$

Such q_1 exists because $\sum_{k=1}^{\infty} \beta_k = -\infty$ by [item 2\)](#) of [Claim 3.10](#).

- Let p_2 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k + \sum_{k=p_1+1}^{p_2} \lambda_k > \alpha.$$

Again, p_2 exists because $\sum \lambda_k = +\infty$.

- Let q_2 be the smallest natural number such that

$$\sum_{k=1}^{p_1} \lambda_k + \sum_{k=1}^{q_1} \beta_k + \sum_{k=p_1+1}^{p_2} \lambda_k + \sum_{k=q_1+1}^{q_2} \beta_k < \alpha.$$

Again, q_2 exists because $\sum \beta_k = -\infty$.

Continuing inductively, we obtain the sequence

$$\lambda_1, \dots, \lambda_{p_1}, \beta_1, \dots, \beta_{q_1}, \lambda_{p_1+1}, \dots, \lambda_{p_2}, \beta_{q_1+1}, \dots, \beta_{q_2}, \lambda_{p_2+1}, \dots \quad (3)$$

whose partial sums alternately pass above and below α .

The picture below illustrates the construction schematically.

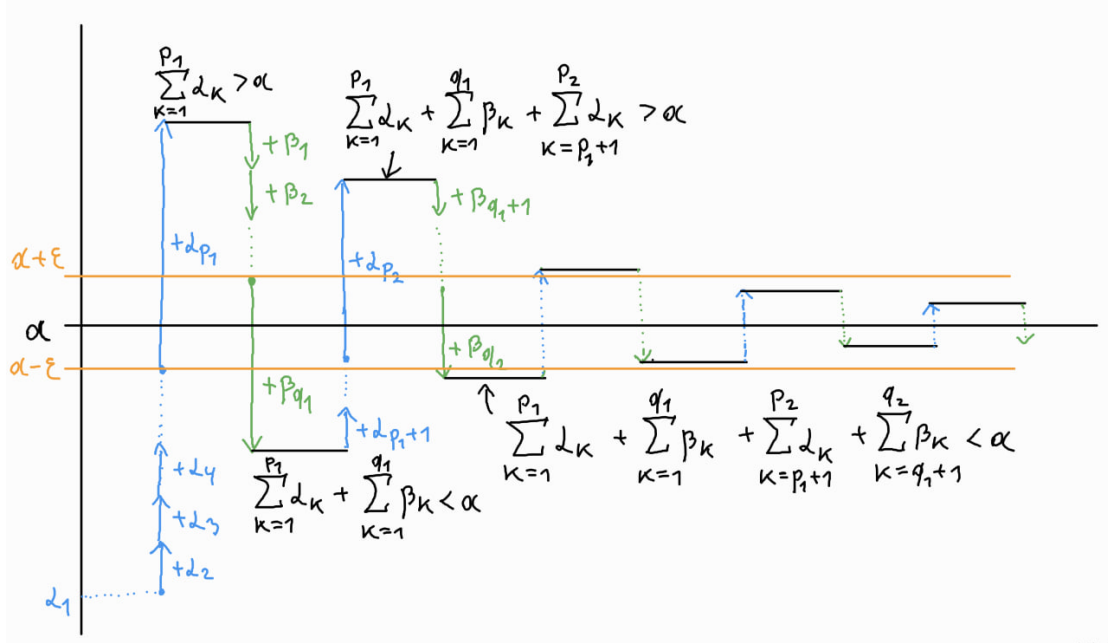


Figure 3.1

The constructed series converges to α . Indeed, note that

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \implies \left(\lim_{k \rightarrow +\infty} a_k = 0 \right) \iff \left(\lim_{k \rightarrow +\infty} \lambda_k = 0 \text{ and } \lim_{k \rightarrow +\infty} \beta_k = 0 \right).$$

Equivalently,

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall k > N, \quad \lambda_k < \varepsilon \quad \text{and} \quad |\beta_k| < \varepsilon.$$

Therefore, after all λ_k and β_k with $k \in \{1, \dots, N\}$ have been included in a partial sum of [sequence \(3\)](#)—which eventually happens because at least one λ_i and one β_i are added at each step—all further deviations of partial sums from α do not exceed ε . This follows from $\lambda_k < \varepsilon$ and $|\beta_k| < \varepsilon$ for $k > N$, together with the choice of the integers p_i and q_i . Due to [Claim 1.2](#) (Cauchy's criterion), the constructed series converges to α .

Every entry in the original sequence $\{a_k\}_{k=1}^{\infty}$ appears in [sequence \(3\)](#) exactly once. Thus [sequence \(3\)](#) is a permutation of the original sequence $\{a_k\}_{k=1}^{\infty}$. $\square$